A Bayesian Transition Model for Clonal T-Cell Dynamics across Tissue and Phenotype in Glioblastoma

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1 Setup

We observe annotated T cells from multiple patients, anatomical sites, and study timepoints. Each observed cell carries five labels: a patient, a TCR clonotype, a study timepoint, an anatomical site, and a discrete T-cell phenotype state.

The data are represented as aggregate clone-by-timepoint-by-site-by-phenotype counts, from which we construct clone-level observations across forward study intervals.

1.1 Indices

We use the following indices throughout.

- Patients: $i = 1, \ldots, I$.
- TCR clonotypes within patient i: $c = 1, \ldots, C_i$.
- Shared study timepoints: $r \in \{1, \ldots, R\}$, ordered as $\tau_1 < \tau_2 < \cdots < \tau_R$. The timepoint labels are comparable across patients.
- Anatomical site: $s \in \mathcal{S}$, with $\mathcal{S} = \{\text{PBMC}, \text{CSF}, \text{Tumor}\}$.
- Annotated T-cell phenotype state: $k \in \{1, \ldots, K\}$. In the current application $K = 11$, and all phenotypes share a single phenotype space.

1.2 Observed count tensor

For each patient i, clonotype c, timepoint r, site s, and phenotype k, let

$$Y_{icrsk} \in \mathbb{Z}_{\geq 0}$$

be the observed number of cells assigned to that combination: how many cells from patient i, clonotype c, and timepoint r were sampled from site s and assigned phenotype k. Collecting the site and phenotype axes,

$$Y_{icr} \in \mathbb{Z}_{\geq 0}^{|\mathcal{S}| \times K}$$

is the tissue-by-phenotype count tensor for clonotype c in patient i at timepoint r.

1.3 Joint tissue–phenotype state space

The model operates on the joint tissue–phenotype state space

$$\mathcal{Z} = \mathcal{S} \times \{1, \dots, K\}, \qquad z = (s, k),$$

where s is the tissue and k is the phenotype. Since $|\mathcal{S}| = 3$ and $K = 11$, the number of joint states is

$$L \;=\; |\mathcal{Z}| \;=\; |\mathcal{S}|\, K \;=\; 33.$$

The model therefore tracks clone counts across a 33-state tissue–phenotype space. When describing transitions we write the source state as $z = (a, u)$ and the destination state as $z' = (b, v)$, where $a, b \in \mathcal{S}$ are source and destination tissues and $u, v \in \{1, \dots, K\}$ are source and destination phenotypes.

1.4 Valid forward intervals

Forward intervals are defined per patient. For patient i, let Q_i be the set of valid forward intervals observed for that patient. Each element $q \in Q_i$ is a pair of study timepoints with source timepoint r_{iq}^{src} and destination timepoint r_{iq}^{dst}. For adjacent study intervals, $r_{iq}^{\mathrm{dst}} = r_{iq}^{\mathrm{src}} + 1$. Here q is not a matrix index; it simply labels a forward interval within patient i.

Each $q \in Q_i$ is one observed forward interval. Intervals such as $T_1 \to T_2$, $T_2 \to T_3$, and $T_3 \to T_4$ are treated as repeated observations of the *same* forward transition process: the transition matrix is shared across all patients and intervals, and the indices i and q merely organize observations rather than selecting a separate matrix.

1.5 Clone-level observations within an interval

Fix a patient i, a valid interval $q \in Q_i$, and a clonotype c observed for that patient. The source and destination count vectors over the joint state space are

$$x_{iqc}(s, k) = Y_{i\,c\,r_{iq}^{\mathrm{src}}\,s\,k}, \qquad y_{iqc}(s, k) = Y_{i\,c\,r_{iq}^{\mathrm{dst}}\,s\,k}, \qquad (s, k) \in \mathcal{Z},$$

so that $x_{iqc} \in \mathbb{Z}_{\geq 0}^{L}$ collects the source counts at the source timepoint and $y_{iqc} \in \mathbb{Z}_{\geq 0}^{L}$ collects the destination counts at the destination timepoint. Their totals are

$$N_{iqc}^{\mathrm{src}} = \sum_{z \in \mathcal{Z}} x_{iqc}(z), \qquad N_{iqc}^{\mathrm{dst}} = \sum_{z \in \mathcal{Z}} y_{iqc}(z).$$

For each patient i and interval $q \in \mathcal{Q}_i$ we keep only clonotypes seen at both ends:

$$\mathcal{C}_{iq} = \left\{ c \in \{1, \ldots, C_i\} \ : \ N^{\mathrm{src}}_{iqc} > 0 \text{ and } N^{\mathrm{dst}}_{iqc} > 0 \right\}.$$

This makes the data hierarchy explicit: patients contain valid forward intervals, and each forward interval has a retained set of clonotypes,

$$i \ \longrightarrow \ q \in \mathcal{Q}_i \ \longrightarrow \ c \in \mathcal{C}_{iq}.$$

A *clone-level forward observation* is therefore a triple (i, q, c) with $q \in \mathcal{Q}_i$ and $c \in \mathcal{C}_{iq}$. For compact notation we collapse the triple into a single index,

$$j \ \equiv \ (i, q, c), \qquad x_j = x_{iqc}, \qquad y_j = y_{iqc},$$

and each observation contributes the tuple $(x_j, y_j, i_j, q_j, c_j)$. Later sections write j for the triple (i, q, c) so the equations stay compact, while the plate diagram (Figure 1) shows the full hierarchy.

1.6 Observed zeros and valid samples

Each patient's interval set $\mathcal{Q}_i$ is built from valid forward intervals that carry the VDJ-profiled samples needed to form source and destination count vectors. Within a valid sample, a zero entry means the clonotype was not observed in that tissue–phenotype state. Implementation concerns—missing VDJ samples, sample-depth filters, and clone dropout—are handled separately from this mathematical setup.

1.7 Setup summary

The setup produces the clone-level forward observations

$$\mathcal{D}_0 = \left\{ (x_j, y_j, i_j, q_j, c_j) \right\}_{j=1}^{J}, \qquad j \equiv (i_j, q_j, c_j),$$

each containing a source count vector, a destination count vector, a patient, a forward interval, and a clonotype identity. The count vectors live in the joint tissue–phenotype state space $\mathcal{Z} = \mathcal{S} \times \{1, \ldots, K\}$, with $|\mathcal{Z}| = 33$.

1.8 Notation summary

Table 1 collects the symbols used throughout.

2 Empirical Source and Destination Summaries

Each clone-level observation $j = (i_j, q_j, c_j)$ comes with two empirical count vectors,

$$x_j, \ y_j \in \mathbb{Z}^{L}_{\geq 0}, \qquad L = 33,$$

where x_j holds the source tissue–phenotype counts for clonotype c_j in patient i_j at the source timepoint of interval q_j, and y_j holds the destination counts for the same clonotype at the destination timepoint.

Symbol	Meaning
$i = 1, \dots, I$	patient index
$c = 1, \dots, C_i$	TCR clonotype within patient i
$r \in \{1, \dots, R\}$	shared study timepoint, $\tau_1 < \cdots < \tau_R$
$s \in \mathcal{S}$	anatomical site, $\mathcal{S} = \{\text{PBMC}, \text{CSF}, \text{Tumor}\}$
$k \in \{1, \dots, K\}$	phenotype state ($K = 11$)
$z = (s, k) \in \mathcal{Z}$	joint tissue–phenotype state ($L = \lvert\mathcal{Z}\rvert = 33$)
Y_{icrsk}	observed count tensor
$\mathcal{Q}_i$	valid forward intervals for patient i
$q \in \mathcal{Q}_i$	forward interval, timepoints $r_{iq}^{\text{src}}, r_{iq}^{\text{dst}}$
$\mathcal{C}_{iq}$	clonotypes present at both ends of interval q
$j = (i_j, q_j, c_j)$	clone-level observation, $j = 1, \dots, J$
$x_j,\ y_j \in \mathbb{Z}_{\geq 0}^{L}$	source / destination count vectors
$N_j^{\text{src}},\ N_j^{\text{dst}}$	source / destination totals
$\theta_j \in \Delta^{L-1}$	empirical source probability vector
$\widehat{\phi}_j$	empirical destination probability vector
$T \in \mathbb{R}^{L \times L}$	shared row-stochastic transition matrix
$\pi_j = \theta_j T$	model-implied destination probability vector
α_z	Dirichlet concentration for row z
$w_{jzz'}$	backward attribution probability
$A_{jzz'}$	latent allocation count
$\mathcal{D}$	full dataset

Table 1: Notation used throughout the model.

2.1 Source and destination totals

The totals

$$N_j^{\text{src}} = \sum_{z \in \mathcal{Z}} x_j(z), \qquad N_j^{\text{dst}} = \sum_{z \in \mathcal{Z}} y_j(z)$$

are both strictly positive, because $c_j \in \mathcal{C}_{q_j}$ guarantees the clonotype is observed at both ends of the interval.

2.2 Source probability vector

Normalizing the source counts gives the empirical source probability vector $\theta_j \in \Delta^{L-1}$, with entries

$$\theta_j(z) = \frac{x_j(z)}{N_j^{\text{src}}}, \qquad z = (s, k).$$

Thus $\theta_j(z)$ is the fraction of observed source cells from clonotype c_j in tissue–phenotype state z. It is a genuine probability vector,

$$\theta_j(z) \geq 0, \qquad \sum_{z \in \mathcal{Z}} \theta_j(z) = 1,$$

and we treat it as a row vector.

2.3 Destination probability vector

The destination counts can likewise be normalized into an empirical destination probability vector $\widehat{\phi}_j \in \Delta^{L-1}$,

$$\widehat{\phi}_j(z) = \frac{y_j(z)}{N_j^{\mathrm{dst}}}.$$

This vector is convenient for descriptive summaries and model checking. The model itself, however, uses the destination *counts* y_j together with the total N_j^{dst}: keeping the destination as counts preserves the distinction between observations supported by many cells and those supported by only a few.

2.4 Optional source smoothing

For sparse clonotypes, the source probability vector can be stabilized by adding a pseudo-count $\delta > 0$ before normalizing:

$$\theta_j^{(\delta)}(z) = \frac{x_j(z) + \delta}{\sum_{z' \in \mathcal{Z}} x_j(z') + L\delta}.$$

Setting $\delta = 0$ recovers the empirical source probability vector θ_j.

3 Joint Tissue–Phenotype Transition Matrix

The central latent object is a single joint tissue–phenotype transition matrix

$$T \in \mathbb{R}^{L \times L}, \qquad L = 33,$$

whose rows are source tissue–phenotype states and whose columns are destination tissue–phenotype states. For source state $z = (a, u)$ and destination state $z' = (b, v)$, the entry

$$T_{zz'} = T_{(a,u),(b,v)} = \Pr\big(\text{tissue } b, \text{ phenotype } v \;\big|\; \text{tissue } a, \text{ phenotype } u\big)$$

is the probability that source state (a, u) contributes to destination state (b, v) over one forward interval.

3.1 Row-stochastic structure

Each row of T is a probability vector over destination states. Hence $T_{zz'} \geq 0$ for all z, z', and every row sums to one:

$$\sum_{z' \in \mathcal{Z}} T_{zz'} = 1 \qquad \Longleftrightarrow \qquad \sum_{b \in \mathcal{S}} \sum_{v=1}^{K} T_{(a,u),(b,v)} = 1$$

for every source state $z = (a, u)$.

3.2 Expected destination probability vector

For observation j with empirical source probability vector θ_j, the model-implied destination probability vector is the pushforward of θ_j through T:

Central transition calculation

$$\pi_j = \theta_j T \in \Delta^{L-1}, \qquad \text{equivalently} \qquad \pi_j(b,v) = \sum_{a\in\mathcal{S}} \sum_{u=1}^{K} \theta_j(a,u)\, T_{(a,u),(b,v)}.$$

The destination probability for state (b,v) is a weighted sum of transition probabilities out of every source state (a,u), weighted by the empirical source distribution.

The corresponding model-implied mean destination count vector is

$$\mathbb{E}\big[y_j \mid T\big] = N_j^{\text{dst}}\, \pi_j.$$

Here y_j is a count vector, π_j is a probability vector, and $N_j^{\text{dst}}\pi_j$ is the expected destination count vector.

3.3 Role of forward intervals

The same matrix T is used for every patient i and interval $q \in \mathcal{Q}_i$. The pair (i,q) identifies a patient and a pair of study timepoints—telling us which source and destination count vectors to build—but it does not select a different transition matrix. All intervals are treated as repeated observations of a common one-step forward transition process.

4 Bayesian Reading of the Transition Matrix

The matrix T is a latent conditional probability kernel over the joint tissue–phenotype state space, and each row $T_{z\cdot}$ is an unknown categorical distribution over destination states.

4.1 Forward generative reading

For observation j, the source probability vector θ_j acts as a mixing distribution over latent source states. Conceptually, for one unit of destination cell mass we first draw a latent source state and then a destination state conditional on it:

$$Z_{\text{src}} \sim \text{Categorical}(\theta_j), \qquad Z_{\text{dst}} \mid Z_{\text{src}} = z,\, T \;\sim\; \text{Categorical}(T_{z\cdot}).$$

Marginalizing over the latent source state reproduces the pushforward:

$$\Pr(Z_{\text{dst}} = z' \mid \theta_j, T) = \sum_{z\in\mathcal{Z}} \theta_j(z)\, T_{zz'} = \pi_j(z'), \qquad \text{so} \qquad \pi_j = \theta_j T.$$

4.2 Likelihood for destination counts

The observed destination data are counts over the 33-state space. Given π_j, they are modeled as multinomial draws:

Observation likelihood

$$y_j \sim \mathrm{Multinomial}\big(N_j^{\mathrm{dst}},\ \pi_j\big), \qquad \pi_j = \theta_j T.$$

The latent transition matrix enters the likelihood only through the pushforward $\pi_j = \theta_j T$.

4.3 Prior over transition rows

Because each row of T is a probability vector, a natural prior is a Dirichlet distribution per row:

$$T_{z\cdot} \sim \mathrm{Dirichlet}(\alpha_z),$$

where α_z collects the prior concentration parameters for the destination distribution of source state z.

Remark (Support restrictions are an implementation detail)**.** In practice the implementation may exclude biologically implausible transitions—for example large backward jumps along a differentiation or activation axis—by fixing the corresponding entries of T to zero, i.e. by placing the Dirichlet only on the retained entries of each row. This sparsifies T and shrinks the parameter count, but it does not change the form of the model: every row remains a Dirichlet-distributed categorical over its destinations, and the expressions below hold verbatim once each sum over destination states is read as a sum over the retained destinations. We therefore treat the support as a fixed preprocessing choice and write $T_{z\cdot} \sim \mathrm{Dirichlet}(\alpha_z)$ over the full destination set throughout.

4.4 Posterior over transition matrices

After observing the data

$$\mathcal{D} = \{(x_j, y_j, i_j, q_j, c_j)\}_{j=1}^{J},$$

inference targets the posterior $p(T \mid \mathcal{D})$, which captures uncertainty about the joint tissue–phenotype transition kernel after all clone-level forward observations. Posterior summaries include posterior means $\mathbb{E}[T_{zz'} \mid \mathcal{D}]$ and credible intervals for individual transition probabilities.

4.5 Posterior predictive destination probabilities

Because T is latent, the implied destination probability vector $\pi_j = \theta_j T$ is also uncertain. The posterior predictive probability of destination state z' is

$$\Pr(Z_{\mathrm{dst}} = z' \mid \theta_j, \mathcal{D}) = \mathbb{E}\big[\pi_j(z') \mid \mathcal{D}\big] = \mathbb{E}\left[\sum_{z \in \mathcal{Z}} \theta_j(z)\, T_{zz'} \;\middle|\; \mathcal{D}\right].$$

4.6 Backward attribution

The transition matrix is oriented forward, $\Pr(Z_{\mathrm{dst}} = z' \mid Z_{\mathrm{src}} = z, T) = T_{zz'}$. The reverse question—which source state likely contributed to an observed destination state—is a posterior attribution. For observation j define

$$w_{jzz'} = \Pr(Z_{\text{src}} = z \mid Z_{\text{dst}} = z',\, \theta_j,\, T).$$

By Bayes' rule, and recognizing the normalizing constant as $\pi_j(z')$,

$$w_{jzz'} = \frac{\theta_j(z)\, T_{zz'}}{\sum_{\tilde{z}\in\mathcal{Z}} \theta_j(\tilde{z})\, T_{\tilde{z}z'}} = \frac{\theta_j(z)\, T_{zz'}}{\pi_j(z')}.$$

This attributes destination state z' back to source state z, conditional on T and the observed source probability vector.

4.7 Latent allocation counts

Since the destination data are counts, the attribution probabilities also define latent allocation counts. Let $A_{jzz'}$ be the unobserved number of destination cells in state z' attributed to latent source state z, so that

$$\sum_{z\in\mathcal{Z}} A_{jzz'} = y_j(z').$$

Conditional on T, θ_j, and $y_j(z')$, these counts are multinomial:

$$\big(A_{j1z'}, \ldots, A_{jLz'}\big) \sim \text{Multinomial}\big(y_j(z'),\, w_{j1z'}, \ldots, w_{jLz'}\big), \qquad L = 33,$$

giving a probabilistic decomposition of observed destination counts into contributions from latent source tissue–phenotype states.

5 Full Generative Story

The full model is a model for the one-interval redistribution of clone mass across tissue–phenotype states. It can be summarized as follows.

Generative model

1. **State space.** Define $\mathcal{Z} = \mathcal{S} \times \{1, \ldots, K\}$ with $|\mathcal{Z}| = 33$. Each state $z = (s, k)$ pairs an anatomical tissue with an annotated T-cell phenotype.
2. **Transition rows.** For each source state z, draw the row

   $$T_{z\cdot} \sim \text{Dirichlet}\big(\alpha_z\big).$$

   Row $T_{z\cdot}$ describes how clone mass starting in state z is redistributed across destinations after one forward interval.
3. **Source distribution.** For each patient i, interval $q \in Q_i$, and retained clonotype $c \in C_{iq}$, form the observation $j = (i, q, c)$ and compute the empirical source probability vector θ_j from x_j. This is where the clone mass sits at the source timepoint.
4. **Pushforward.** Map the source distribution forward, $\pi_j = \theta_j T$ — where the model expects the clone mass to appear at the destination timepoint.
5. **Observation.** Draw the destination counts

   $$y_j \sim \text{Multinomial}\big(N_j^{\text{dst}},\, \pi_j\big).$$

In words: each clone begins an interval with an observed distribution across tissue–phenotype states; the transition matrix maps that source distribution forward into a destination probability vector; and the observed destination counts are sampled from that vector, with the destination total N_j^{dst} controlling how much count information the observation contributes.

Putting the pieces together, the generative model is, compactly,

$$T_{z\cdot} \sim \mathrm{Dirichlet}(\alpha_z), \qquad \pi_j = \theta_j T, \qquad y_j \sim \mathrm{Multinomial}(N_j^{\mathrm{dst}}, \pi_j),$$

with the corresponding dependency structure shown in Figure 1.

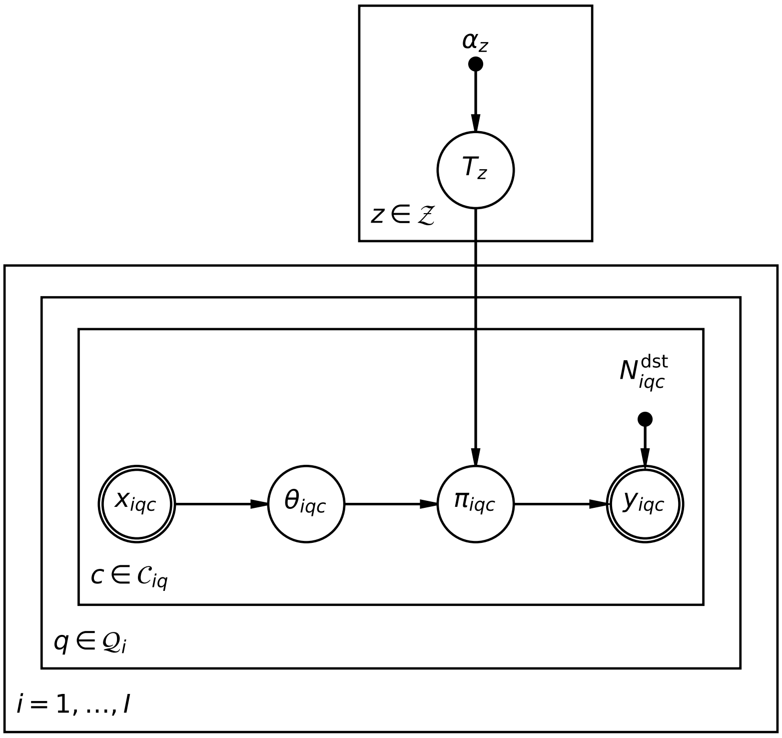

Figure 1: Plate diagram of the generative model. The transition rows $T_{z\cdot}$ are drawn once per source state z (top plate, indexed by $z \in \mathcal{Z}$) from a Dirichlet prior with concentration α_z. This transition block sits *outside* the patient plate: a single matrix T is shared across all patients and intervals. The observation side is drawn with explicit nesting—patient i, forward interval $q \in \mathcal{Q}_i$, retained clonotype $c \in \mathcal{C}_{iq}$—so a clone-level observation is the triple (i, q, c). Within the innermost plate each observation contributes the empirical source distribution θ_{iqc}, deterministically normalized from the observed source counts x_{iqc}; the pushforward $\pi_{iqc} = \theta_{iqc} T$ then drives a multinomial draw of the observed destination counts y_{iqc} with total N_{iqc}^{dst}. Double circles (x_{iqc}, y_{iqc}) mark observed quantities and solid dots (α_z, N_{iqc}^{dst}) mark fixed inputs. The latent allocation counts used for fitting are a computational augmentation and are introduced separately in Section 7.

6 Inference

Inference estimates the posterior over the shared transition matrix T. The observed data are the clone-level forward observations

$$\mathcal{D} = \left\{(x_j, y_j, i_j, q_j, c_j)\right\}_{j=1}^{J}.$$

The source probability vectors θ_j and destination totals N_j^{dst} are derived from the observed count vectors, while the prior parameters α are fixed model inputs.

6.1 Learned quantity

The main learned object is the posterior $p(T \mid \mathcal{D})$, whose summaries describe the inferred one-interval redistribution of clone mass across tissue–phenotype states. A single-transition summary is

$$\mathbb{E}\left[T_{(a,u),(b,v)} \mid \mathcal{D}\right],$$

the posterior mean probability that source state (a, u) contributes to destination state (b, v) over one forward interval.

6.2 Derived tissue-level summaries

Because T lives on tissue–phenotype states, tissue-level summaries follow by summing over phenotypes. These are not separate parameters but functions of the posterior over T; their purpose is to compress the 33×33 matrix into interpretable tissue-level movement. For a source tissue a, source phenotype u, and destination tissue b,

$$m_{a,u\to b} = \sum_{v=1}^{K} T_{(a,u),(b,v)}$$

is the probability that source state (a, u) maps into destination tissue b, marginalizing over destination phenotype. Averaging over a source phenotype distribution $\rho_a(u)$ gives a phenotype-averaged tissue summary,

$$m_{a\to b} = \sum_{u=1}^{K} \rho_a(u) \sum_{v=1}^{K} T_{(a,u),(b,v)},$$

where ρ_a may be chosen empirically—for example, the observed phenotype distribution among source cells in tissue a.

6.3 Phenotype summaries conditional on destination tissue

For a fixed source tissue a, destination tissue b, and source phenotype u, define

$$h_{a,u\to b}(v) = \frac{T_{(a,u),(b,v)}}{\sum_{v'=1}^{K} T_{(a,u),(b,v')}},$$

the destination phenotype distribution *conditional* on ending in tissue b. Together, the two summaries separate two questions:

$$\underbrace{m_{a,u\to b}}_{\text{how much mass reaches tissue } b} \qquad \text{versus} \qquad \underbrace{h_{a,u\to b}(v)}_{\text{which phenotypes appear within tissue } b}.$$

6.4 Posterior predictive checks

For each observation j, posterior draws of T induce posterior draws of $\pi_j = \theta_j T$, which can be used to simulate replicated destination counts

$$y_j^{\mathrm{rep}} \sim \mathrm{Multinomial}\big(N_j^{\mathrm{dst}}, \pi_j\big).$$

Comparing y_j^{rep} to the observed y_j checks whether the inferred transition matrix can reproduce observed destination tissue–phenotype count patterns.

6.5 Posterior attribution summaries

For each observation j, the posterior attribution probabilities

$$w_{jzz'} = \frac{\theta_j(z)\, T_{zz'}}{\pi_j(z')}$$

attribute observed destination states back to source states. They answer questions such as: *given destination tumor-exhausted cells, which source tissue–phenotype states most likely contributed to them?* Attribution is computed from T, θ_j, and π_j, and can be aggregated across observations, tissues, phenotypes, or tissue blocks.

7 Computational Inference

This section reuses the generative model of Section 5 (summarized in Figure 1) and describes how its posterior target $p(T \mid \mathcal{D})$ is approximated and how the fitted transition matrix is obtained. The objects are exactly those in the plate diagram: the shared transition matrix T over the joint tissue–phenotype state space $\mathcal{Z} = \mathcal{S} \times \{1, \ldots, K\}$ ($|\mathcal{Z}| = L = 33$), the empirical source distributions θ_j, the pushforwards $\pi_j = \theta_j T$, and the observed destination counts y_j.

Recall the data $\mathcal{D} = \{(x_j, y_j, i_j, q_j, c_j)\}_{j=1}^{J}$. For each observation j, the source counts x_j are normalized into θ_j, and the destination counts are modeled as

$$y_j \sim \mathrm{Multinomial}(N_j^{\mathrm{dst}}, \pi_j), \qquad \pi_j = \theta_j T, \qquad \pi_j(z') = \sum_{z \in \mathcal{Z}} \theta_j(z)\, T_{zz'}.$$

The difficulty is that an observed destination count in state z' could have been contributed by *any* source state z, and these source-to-destination allocations are unobserved. To make the posterior tractable, variational inference *augments* the generative model of Figure 1 with latent allocation counts—variables that do not appear in the generative story itself—and then fits a factorized approximation by coordinate ascent.

7.1 Latent allocation counts

Let $\xi_{jzz'}$ be the unobserved number of destination cells in state z' for observation j attributed to source state z. By construction they decompose each observed destination count,

$$\sum_{z \in \mathcal{Z}} \xi_{jzz'} = y_j(z').$$

These are computational latent variables that split the observed destination vector into unobserved source-to-destination contributions. Given T and θ_j, the exact posterior attribution probability of source state z for destination state z' is the backward attribution quantity from Section 4.6,

$$w_{jzz'} = \frac{\theta_j(z)\, T_{zz'}}{\sum_{\tilde{z}\in\mathcal{Z}} \theta_j(\tilde{z})\, T_{\tilde{z}z'}}.$$

Variational inference replaces these with an approximate responsibility derived from the variational distribution.

7.2 Variational family

To avoid overloading the interval index $q \in Q_i$, we write the variational distribution as $\mathfrak{q}(T, \xi)$ and adopt a mean-field factorization

$$\mathfrak{q}(T, \xi) = \mathfrak{q}(T)\, \mathfrak{q}(\xi).$$

The transition matrix receives independent Dirichlet factors, one per row,

$$\mathfrak{q}(T) = \prod_{z\in\mathcal{Z}} \mathrm{Dirichlet}\big(T_{z\cdot};\, \lambda_{z\cdot}\big),$$

and the allocation variables receive multinomial factors,

$$\mathfrak{q}(\xi) = \prod_{j=1}^{J} \prod_{z'\in\mathcal{Z}} \mathrm{Multinomial}\big(\xi_{j\cdot z'};\, y_j(z'),\, r_{j\cdot z'}\big),$$

where $r_{jzz'}$ is the variational responsibility assigning destination state z' in observation j to source state z.

7.3 Coordinate updates

The algorithm alternates between updating the allocation responsibilities $r_{jzz'}$ and the Dirichlet parameters $\lambda_{zz'}$.

7.3.1 Allocation update

For each observation j, source state z, and destination state z',

$$r_{jzz'} \propto \theta_j(z)\, \exp\!\big(\mathbb{E}_{\mathfrak{q}}[\log T_{zz'}]\big),$$

normalized over source states so that $\sum_{z\in\mathcal{Z}} r_{jzz'} = 1$ for each j and z'. Under the Dirichlet factor, the required expectation is

$$\mathbb{E}_{\mathfrak{q}}[\log T_{zz'}] = \psi(\lambda_{zz'}) - \psi\Big(\sum_{\tilde{z}\in\mathcal{Z}} \lambda_{z\tilde{z}}\Big),$$

where ψ is the digamma function. Biologically, this step probabilistically assigns observed destination tissue–phenotype counts back to the source tissue–phenotype states that could have produced them.

7.3.2 Transition-row update

For each source state z and destination state z',

$$\lambda_{zz'} = \alpha_{zz'} + \sum_{j=1}^{J} y_j(z')\, r_{jzz'},$$

so the variational posterior parameter is the prior concentration plus the expected number of transitions allocated from source state z to destination state z'.

7.4 Optimization objective

The coordinate updates maximize the evidence lower bound (ELBO)

$$\mathcal{L}(\mathfrak{q}) = \mathbb{E}_{\mathfrak{q}}[\log p(T, \xi, \mathcal{D})] - \mathbb{E}_{\mathfrak{q}}[\log \mathfrak{q}(T, \xi)]\,,$$

and the algorithm iterates between the allocation update and the transition-row update until $\mathcal{L}(\mathfrak{q})$ stabilizes.

7.5 Estimated transition matrix

After convergence, the fitted transition matrix is summarized by the variational posterior mean $\widehat{T}_{zz'} = \mathbb{E}_{\mathfrak{q}}[T_{zz'}]$, the normalized Dirichlet parameter:

$$\widehat{T}_{zz'} = \frac{\lambda_{zz'}}{\sum_{\bar{z} \in \mathcal{Z}} \lambda_{z\bar{z}}}.$$

The matrix $\widehat{T}$ is the estimated one-interval transition probability matrix over the 33-state tissue–phenotype space, and it is the fitted quantity used for posterior summaries, posterior predictive checks, posterior attribution summaries, and subsequent rate decomposition.

7.6 Algorithm summary

Variational algorithm (CAVI)

1. **Initialize** the variational Dirichlet parameters λ.
2. **Repeat until the ELBO $\mathcal{L}(\mathfrak{q})$ converges:**
   - update the allocation responsibilities $r_{jzz'} \propto \theta_j(z)\, \exp\big(\mathbb{E}_{\mathfrak{q}}[\log T_{zz'}]\big)$;
   - update the transition-row parameters $\lambda_{zz'} = \alpha_{zz'} + \sum_j y_j(z')\, r_{jzz'}$.
3. **Return** the posterior mean $\widehat{T} = \mathbb{E}_{\mathfrak{q}}[T]$.

The responsibility update estimates which source tissue–phenotype states explain the observed destination counts; the transition-row update accumulates those expected allocations into a posterior estimate of the shared forward transition matrix.